

Geometry of Data HW-4

Sruvanth Chaudhary Potluri

UND50V

Problem ①

Part (a)

fisher information metric $g = \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{2}{\sigma^2} \end{pmatrix} = g_{ij}$ for (μ, σ)

$$g^{ij} = g_{ij}^{-1} = \frac{\text{Adj}(g)}{|g|} = \frac{\begin{pmatrix} \frac{2}{\sigma^2} & 0 \\ 0 & \frac{1}{\sigma^2} \end{pmatrix}}{\frac{2}{\sigma^4}} = \begin{pmatrix} \sigma^2 & 0 \\ 0 & \frac{\sigma^2}{2} \end{pmatrix}$$

Christoffel symbols $\Gamma_{ij}^k = \frac{1}{2} \sum_{l=1}^n g^{kl} \left(\frac{\partial g_{jl}}{\partial x^i} + \frac{\partial g_{il}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^l} \right)$ $x^1 = \mu$
 $x^2 = \sigma$

the non-zero g^{kl} are $g^{11} = \sigma^2$, $g^{22} = \frac{\sigma^2}{2}$

the non zero partial derivatives of the metric components are

$$\frac{\partial g_{11}}{\partial \sigma} = \frac{-2}{\sigma^3} \quad \frac{\partial g_{22}}{\partial \sigma} = \frac{-4}{\sigma^3}$$

finding the Christoffel symbols $\rightarrow 0$

$$\Gamma_{11}^1 = \frac{1}{2} \left(g^{11} \left(\frac{\partial g_{11}}{\partial x^1} + \frac{\partial g_{11}}{\partial x^1} - \frac{\partial g_{11}}{\partial x^1} \right) + g^{12} \left(\frac{\partial g_{11}}{\partial x^2} + \frac{\partial g_{11}}{\partial x^2} - \frac{\partial g_{11}}{\partial x^2} \right) \right)$$
$$= \frac{1}{2} \left(g^{11} (0 + 0 + 0) \right) = 0$$

$$\begin{aligned}
 \Gamma_{12}' = \Gamma_{21}' &= \frac{1}{2} \left(g''(\cdot) + g'^2(\cdot) \right) \\
 &= \frac{1}{2} \left(g''(0 + \frac{\partial g_{11}}{\partial x^2} - 0) \right) \\
 &= \frac{1}{2} \times \sigma^2 \times -\frac{2}{\sigma^3} \\
 &= -\frac{1}{\sigma}
 \end{aligned}$$

$$\begin{aligned}
 \Gamma_{22}' &= \frac{1}{2} \left(g''(\cdot) + g'^2(\cdot) \right) \\
 &= \frac{1}{2} \left(g''(0 + 0 - 0) \right) = 0
 \end{aligned}$$

$$\begin{aligned}
 \Gamma_{11}^2 &= \frac{1}{2} \left(g_{\downarrow 0}^{21}(\cdot) + g^{22}(\cdot) \right) \\
 &= \frac{1}{2} g^{22} \left(0 + 0 - \frac{\partial g_{11}}{\partial x^2} \right) \\
 &= \frac{1}{2} \cdot \frac{\sigma^2}{2} \cdot \frac{+2}{\sigma^3} = \frac{1}{2\sigma}
 \end{aligned}$$

$$\begin{aligned}
 \Gamma_{12}^2 = \Gamma_{21}^2 &= \frac{1}{2} \left(g_{\downarrow 0}^{21}(\cdot) + g^{22}(\cdot) \right) \\
 &= \frac{1}{2} g^{22} (0 + 0 - 0) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}\Gamma_{22}^2 &= \frac{1}{2} \left(g_{\downarrow 0}^{21} () + g^{22} () \right) \\ &= \frac{1}{2} \left(g^{22} \left(\frac{-4}{\sigma^3} + \frac{-4}{\cancel{\sigma^3}} + \frac{4}{\cancel{\sigma^3}} \right) \right) \\ &= \frac{1}{2} \cdot \frac{\sigma^2}{2} \cdot \frac{-4}{\sigma^3} = -\frac{1}{\sigma}\end{aligned}$$

The non-zero Christoffel symbols are

$$\Gamma_{12}^1 = \Gamma_{21}^1 = -\frac{1}{\sigma} \quad \left| \quad \Gamma_{11}^2 = \frac{1}{2\sigma} \quad \right| \quad \Gamma_{22}^2 = -\frac{1}{\sigma}$$

Part (b)

$$(\kappa(t), \sigma(t)) = (a, be^{ct})$$

Verify whether the curves of the following form satisfy the geodesic equation

for a curve to be a geodesic, it has to satisfy the differential equation

$$\frac{d^2 \gamma^k}{dt^2} = - \sum_{i,j=1}^n \Gamma_{ij}^k (\gamma(t)) \frac{d\gamma^i}{dt} \frac{d\gamma^j}{dt} \quad \text{where } i,j = \{1,2\}$$

$$\frac{d\gamma^1}{dt} = 0 \quad \left| \quad \frac{d^2 \gamma^1}{dt^2} = 0 \quad \right| \quad \frac{d\gamma^2}{dt} = bc e^{ct} \quad \left| \quad \frac{d^2 \gamma^2}{dt^2} = bc^2 e^{ct} = c^2 \sigma(t) \right|$$

for $k=1$, the only non-zero Christoffel symbols are Γ_{12}^1 and Γ_{21}^1
 in both of these cases the R.H.S terms have $\frac{d\gamma^1}{dt}$, which is
 equal to zero and the RHS $\frac{d^2\gamma^1}{dt^2}$ is also zero
 and subsequently all the terms are zero

$\therefore$ in the case of $k=1$, the D.E is satisfied

for $k=2$

$$\frac{d^2\gamma^2}{dt^2} = b c^2 e^{ct} \longrightarrow \text{LHS}$$

$$= c^2 \sigma(t)$$

$$\text{RHS} = - \left(\underbrace{\Gamma_{11}^{22}}_{\substack{\downarrow \\ \frac{1}{2\sigma(t)}}} \cdot \underbrace{\frac{d\gamma^1}{dt}}_{\downarrow 0} \cdot \underbrace{\frac{d\gamma^1}{dt}}_{\downarrow 0} + \underbrace{\Gamma_{22}^{22}}_{\substack{\downarrow \\ -\frac{1}{\sigma(t)}}} \cdot \underbrace{\frac{d\gamma^2}{dt}}_{\downarrow c\sigma(t)} \cdot \underbrace{\frac{d\gamma^2}{dt}}_{\downarrow c\sigma(t)} \right)$$

$$= - \left(0 + \left(-\frac{1}{\sigma(t)} \cdot c\sigma(t) \cdot c\sigma(t) \right) \right)$$

$$= c^2 \sigma(t)$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

Since L.H.S = R.H.S

the D.E is satisfied for $k=2$

Since the D.E is satisfied for both $k=1, k=2$, the given equation of curves satisfy the geodesic equation

Part (c)

Verify if
(curves of the form

$$(X(t), \sigma(t)) = \left(a \tanh(ct) + b, a \frac{\sqrt{2}}{2} \operatorname{sech}(ct) \right)$$

satisfy the geodesic equation

Similar to the previous part, they have to satisfy the differential equation

$$\frac{dY'}{dt} = a \operatorname{sech}^2(ct) \cdot c = ac \operatorname{sech}^2(ct)$$

$$\begin{aligned} \frac{d^2 Y'}{dt^2} &= ac \cdot 2 \operatorname{sech}(ct) (-\operatorname{sech} ct \cdot \tanh ct) \cdot c \\ &= -2ac^2 \operatorname{sech}^2(ct) \tanh(ct) \end{aligned}$$

$$\frac{dY''}{dt} = a \frac{\sqrt{2}}{2} (-\operatorname{sech}(ct) \cdot \tanh(ct)) \cdot c = -ac \frac{\sqrt{2}}{2} \operatorname{sech}(ct) \tanh(ct)$$

$$\begin{aligned} \frac{d^2 Y''}{dt^2} &= -ac \frac{\sqrt{2}}{2} \left(\operatorname{sech}(ct) \cdot \operatorname{sech}^2(ct) \cdot c + \tanh(ct) \cdot (-\operatorname{sech}(ct) \cdot \tanh(ct)) \cdot c \right) \\ &= -ac^2 \frac{\sqrt{2}}{2} \left(\operatorname{sech}^3(ct) - \tanh^2(ct) \operatorname{sech}(ct) \right) \end{aligned}$$

for $k=1$

$$LHS = -2ac^2 \operatorname{sech}^2(ct) \tanh(ct)$$

$$RHS = -\left(\Gamma_{12}' \left(\frac{dY^1}{dt} \cdot \frac{dY^2}{dt} \right) + \Gamma_{21}' \left(\frac{dY^2}{dt} \cdot \frac{dY^1}{dt} \right) \right) \quad \text{Since } \Gamma_{12}' = \Gamma_{21}'$$

$$= -\left(2 \cdot \Gamma_{12}' \cdot \frac{dY^1}{dt} \cdot \frac{dY^2}{dt} \right)$$

$$= -\left(2 \cdot \frac{-1}{\sigma(t)} \cdot ac \operatorname{sech}^2(ct) \cdot -ac \frac{\sqrt{2}}{2} \operatorname{sech}(ct) \tanh(ct) \right)$$

$\downarrow$
 $a \frac{\sqrt{2}}{2} \operatorname{sech}(ct)$

$$= -\left(2 \cdot \frac{1}{a \frac{\sqrt{2}}{2} \operatorname{sech}(ct)} \cdot ac \operatorname{sech}^2(ct) \cdot ac \frac{\sqrt{2}}{2} \operatorname{sech}(ct) \tanh(ct) \right)$$

$$= -\left(2 \cdot c \cdot \operatorname{sech}^2(ct) \cdot ac \cdot \tanh(ct) \right)$$

$$= -2ac^2 \operatorname{sech}^2(ct) \cdot \tanh(ct)$$

$$LHS = RHS$$

The family of curves satisfy the differential equation for $k=1$

$$LHS = \frac{d^2 Y^2}{dt^2} = -ac^2 \frac{\sqrt{2}}{2} \left(\operatorname{sech}^3(ct) - \tanh^2(ct) \operatorname{sech}(ct) \right)$$

for $k=2$

$$\begin{aligned}
RHS &= -\left(\Gamma_{11}^2 \cdot \frac{dY^1}{dt} \cdot \frac{dY^1}{dt} + \Gamma_{22}^2 \cdot \frac{dY^2}{dt} \cdot \frac{dY^2}{dt} \right) \\
&= -\left(\frac{1}{2\sigma(t)} \cdot \left(\frac{dY^1}{dt} \right)^2 - \frac{1}{\sigma(t)} \left(\frac{dY^2}{dt} \right)^2 \right) \\
&= -\left(\frac{1}{\sigma(t)} \left(\frac{1}{2} (ac \sec h^2(ct))^2 - \left(-ac \frac{\sqrt{2}}{2} \operatorname{sech}(ct) \tanh(ct) \right)^2 \right) \right) \\
&= \frac{1}{\left(a \frac{\sqrt{2}}{2} \operatorname{sech}(ct) \right)} \cdot \left(\frac{a^2 c^2}{2} \sec h^4(ct) - \frac{a^2 c^2}{2} \sec h^2(ct) \tanh^2(ct) \right) \\
&= \frac{a^2 c^2}{\cancel{2} \cdot \cancel{a} \cdot \frac{\sqrt{2}}{2} \operatorname{sech}(ct)} \cdot \left(\sec h^3(ct) - \sec h(ct) \cdot \tanh^2(ct) \right) \\
&= \underbrace{a c^2 \cdot \frac{\sqrt{2}}{2}}_{\substack{\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}}} \cdot \left(\sec h^3(ct) - \tanh^2(ct) \cdot \sec h(ct) \right) \\
&\xrightarrow{L.H.S = R.H.S}
\end{aligned}$$

the differential equation is satisfied for $k=2$

Therefore since the differential equation is satisfied by the given form of curves for both $k=1$ and $k=2$, The curves represent a geodesic equation

Part (D)

Given :- two gaussians with (μ, σ) $(0, 1)$ and $(0, 2)$

Graph of the PDF's in jupyter notebook

we use the equation $(\mu(t), \sigma(t)) = (a, be^{ct})$ to represent the curves
 $t \in [0, 1]$

So, a curve from $t=0 \rightarrow 1$ is from $(0, 1) \rightarrow (0, 2)$

So to get the constants a, b, c

We know that

$$\begin{aligned} (\mu(0), \sigma(0)) &= (0, 1) = (a, be^{c \cdot 0}) \quad \xrightarrow{t=0} \\ (\mu(1), \sigma(1)) &= (0, 2) = (a, be^{c \cdot 1}) \quad \xrightarrow{t=1} \end{aligned}$$

from these it is clear that $a=0, be^0=1 \Rightarrow \boxed{b=1}$
and we get $e^c=2 \Rightarrow c=\ln 2$

$\therefore$ the geodesic curve between the 2 points is $\begin{pmatrix} 0, e^{t \ln 2} \\ 0, 2^t \end{pmatrix}$

The length of the geodesic on a riemannian manifold with metric tensor g is given by

$$L(\gamma) = \int_a^b \sqrt{g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))} dt$$

in the current scenario

$$g = \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{2}{\sigma^2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2^{2t}} & 0 \\ 0 & \frac{2}{2^{2t}} \end{pmatrix}$$

$$\gamma(t) = (\dot{x}(t), \dot{y}(t)) = (0, \ln 2 \cdot 2^t) = (0, \ln 2 \cdot 2^t)$$

$$L(\gamma) = \int_0^1 \sqrt{\begin{pmatrix} 0 \\ \ln 2 \cdot 2^t \end{pmatrix}^T \begin{pmatrix} \frac{1}{2^{2t}} & 0 \\ 0 & \frac{2}{2^{2t}} \end{pmatrix} \begin{pmatrix} 0 \\ \ln 2 \cdot 2^t \end{pmatrix}} dt$$

$$= \int_0^1 \sqrt{\begin{pmatrix} 0 & \ln 2 \cdot 2^t \cdot \frac{2}{2^{2t}} \end{pmatrix} \begin{pmatrix} 0 \\ \ln 2 \cdot 2^t \end{pmatrix}} dt$$

$$= \int_0^1 \sqrt{\begin{pmatrix} 0 & 2^{1-t} \ln 2 \end{pmatrix} \begin{pmatrix} 0 \\ \ln 2 \cdot 2^t \end{pmatrix}} dt$$

$$= \int_0^1 \sqrt{0 + (\ln 2)^2 \cdot 2} dt$$

$$= \int_0^1 \ln 2 \cdot \sqrt{2} dt = \sqrt{2} \ln 2 \approx 0.98$$

Therefore the geodesic distance between the two gaussians is

$$\sqrt{2} \ln 2$$

The plot for the geodesic is plotted in the jupyter notebook

Now we do the same for $(0, 4)$ and $(0, 8)$

we know that $t \in [0, 1]$

at $t=0$

$$(a, be^0) = (0, 4) \rightarrow a=0, b=4$$

at $t=1$

$$(a, be^c) = (0, 8) \rightarrow 4e^c = 8 \Rightarrow e^c = 2, c = \ln 2$$

our curve equation now becomes $(0, 4e^{t \ln 2})$ or $(0, 4 \cdot 2^t)$

our metric now becomes

$$g = \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{2}{\sigma^2} \end{pmatrix} = \begin{pmatrix} \frac{1}{(4 \cdot 2^t)^2} & 0 \\ 0 & \frac{2}{(4 \cdot 2^t)^2} \end{pmatrix}$$

$$\dot{\gamma}(t) = (\dot{x}(t), \dot{y}(t)) = (0, c \sigma(t)) = (0, \ln 2 \cdot 4 \cdot 2^t)$$

$$L(\gamma) = \int_0^1 \sqrt{\begin{pmatrix} 0 \\ 4 \ln 2 \cdot 2^t \end{pmatrix}^t \begin{pmatrix} \frac{1}{(4 \cdot 2^t)^2} & 0 \\ 0 & \frac{2}{(4 \cdot 2^t)^2} \end{pmatrix} \begin{pmatrix} 0 \\ 4 \ln 2 \cdot 2^t \end{pmatrix}} dt$$

$$= \int_0^1 \sqrt{\begin{pmatrix} 0 & 4 \cdot \ln 2 \cdot 2^t \cdot 2 \\ 4 \cdot 2^t \cdot 4 \cdot 2^t \end{pmatrix} \begin{pmatrix} 0 \\ 4 \ln 2 \cdot 2^t \end{pmatrix}} dt$$

$$= \int_0^1 \sqrt{\frac{\ln 2 \cdot 2 \cdot 4 \cdot \ln 2 \cdot 2^t}{4 \cdot 2^t}} dt = \int_0^1 \sqrt{2} \ln 2 dt$$

$$= \sqrt{2} \ln 2 \approx 0.98$$

Plots in Jupyter notebook

Let us try to find a general formula for distance in this case when the curves are

$$(\mu(t), \sigma(t)) = (a, b e^{ct})$$

$$g = \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{2}{\sigma^2} \end{pmatrix} \quad \gamma(t) = (0, c \sigma(t))$$

$$\begin{aligned} L(\gamma) &= \int_0^1 \sqrt{\begin{pmatrix} 0 \\ c\sigma(t) \end{pmatrix}^T \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{2}{\sigma^2} \end{pmatrix} \begin{pmatrix} 0 \\ c\sigma(t) \end{pmatrix}} dt \\ &= \int_0^1 \sqrt{\begin{pmatrix} 0 & \frac{2c}{\sigma(t)} \end{pmatrix} \begin{pmatrix} 0 \\ c\sigma(t) \end{pmatrix}} dt \\ &= \int_0^1 \sqrt{2c^2} dt \\ &= \int_0^1 c\sqrt{2} dt = c\sqrt{2} [t]_0^1 = c\sqrt{2} \end{aligned}$$

this result is interesting because it shows that the distance is independent of the values of a, b and that since we had $c = \ln 2$ in both the cases in this part we get the same result that is $\sqrt{2} \ln 2$ which is consistent with our calculation of a general distance formula. Please note that this formula is only applicable when the mean of the end points is same. in this scenario the constants a and b don't matter and b acts as a scaling factor which absorbs any equal scaling applied to both the points. Conjecture in next page

from our observations and calculations of the distances so far, the general rule we can conjecture is that the geodesic distance does not change when we scale both the standard deviations by the same factor, when the mean μ is the same for both of the points, this is because in that scenario a doesn't matter and c remains the same and b doesn't matter to the distance
_{because the curve is (a, b, c)} _{scaling factor}

Part (c)

Now, similar to the previous question, we have to find the geodesic curve for $(-1, 1)$ and $(1, 1)$

let us use the equation $(\mu(t), \sigma(t)) = (a \tanh(ct) + b, a \frac{\sqrt{2}}{2} \operatorname{sech}(ct))$
 for $t \in [-1, 1]$

we can get 4 equations from this, they are

$$a \tanh(-c) + b = -1 \quad \text{--- ①} \quad a \frac{\sqrt{2}}{2} \operatorname{sech}(c) = a \frac{\sqrt{2}}{2} \operatorname{sech}(-c) = 1$$

$$a \tanh(c) + b = 1 \quad \text{--- ②}$$

by subtracting ① from ②

$$a \tanh(-c) + b - a \tanh(c) - b = -1 - 1$$

$$-2a \tanh c = -2$$

$$a \tanh c = +1 \Rightarrow b = 0 \text{ if you put it in ②}$$

$$\tanh c = \frac{1}{a} \quad \text{--- ③}$$

$$\boxed{\tanh(-x) = -\tanh(x)}$$

We have

$$\frac{a\sqrt{2}}{2} \operatorname{sech}(c) = 1$$

↓

$$\frac{a\sqrt{2}}{2} \sqrt{1 - \tanh^2 c} = 1$$

from (3)

$$\frac{a\sqrt{2}}{2} \sqrt{1 - \frac{1}{a^2}} = 1$$

$$\frac{a\sqrt{2}}{2} \cdot \frac{1}{a} \cdot \sqrt{a^2 - 1} = 1 \Rightarrow \sqrt{a^2 - 1} = \sqrt{2}$$

$$a^2 - 1 = 2 \Rightarrow \boxed{a = \pm \sqrt{3}}$$

for the sake of simplicity ^{and validity} we use $\boxed{a = \sqrt{3}}$

and $\boxed{c = \tanh^{-1}\left(\frac{1}{\sqrt{3}}\right)}$, now, our curve is $\left(\sqrt{3} \tanh\left(\left(\tanh^{-1}\frac{1}{\sqrt{3}}\right)t\right), \frac{\sqrt{6}}{2} \operatorname{sech}\left(\left(\tanh^{-1}\frac{1}{\sqrt{3}}\right)t\right)\right)$

Geodesic and PDF plot in jupyter notebook

Similar to the previous part, we calculate the distance

the metric we use

$$g = \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{2}{\sigma^2} \end{pmatrix}$$

Our curve is

$$(x(t), y(t)) = \left(\sqrt{3} \tanh\left(\tanh^{-1}\left(\frac{1}{\sqrt{3}}\right) \cdot t\right), \sqrt{\frac{3}{2}} \operatorname{sech}\left(\tanh^{-1}\left(\frac{1}{\sqrt{3}}\right) \cdot t\right) \right)$$

$$(x(t), y(t)) = \left(ac \operatorname{sech}^2(ct), -ac \frac{\sqrt{2}}{2} \operatorname{sech}(ct) \tanh(ct) \right)$$

$$L(Y) = \int_{-1}^1 \sqrt{\begin{pmatrix} ac \operatorname{sech}^2(ct) \\ -ac \frac{\sqrt{2}}{2} \operatorname{sech}(ct) \tanh(ct) \end{pmatrix}^T \begin{pmatrix} \frac{1}{2} \operatorname{sech}^2(ct) & 0 \\ 0 & \frac{2}{a^2 \operatorname{sech}^2(ct)} \end{pmatrix} \cdot \begin{pmatrix} \\ \end{pmatrix}} dt$$

$$L(Y) = \int_{-1}^1 \left(\frac{ac \operatorname{sech}^2(ct)}{\frac{a^2}{2} \operatorname{sech}^2(ct)} \cdot \frac{2 \cdot \left(-ac \frac{\sqrt{2}}{2} \operatorname{sech}(ct) \tanh(ct)\right)}{\frac{a^2}{2} \operatorname{sech}^2(ct)} \right) dt$$

$$= \int_{-1}^1 \left(\frac{2c}{a} - \frac{2\sqrt{2}c \tanh(ct)}{a \cdot \operatorname{sech}(ct)} \right) \begin{pmatrix} ac \operatorname{sech}^2(ct) \\ -ac \frac{\sqrt{2}}{2} \operatorname{sech}(ct) \tanh(ct) \end{pmatrix} dt$$

$$= \int_{-1}^1 2c \operatorname{sech}^2(ct) - \frac{\cancel{2\sqrt{2}c \tanh(ct)}}{\cancel{a \operatorname{sech}(ct)}} \cdot \cancel{-ac \frac{\sqrt{2}}{2} \operatorname{sech}(ct) \tanh(ct)} dt$$

$$= \int_{-1}^1 2c \operatorname{sech}^2(ct) + 2c \tanh^2(ct) dt$$

$$= \int_{-1}^1 2c (\operatorname{sech}^2(ct) + \tanh^2(ct)) dt$$

W.k.t
 $\operatorname{sech}^2(ct) + \tanh^2(ct) = 1$
 from identities

$$= \int_{-1}^1 2c \, dt = 2c [t]_{-1}^1 = 4c$$

$\therefore$ the general formula for distance when the curves are in this format and there is no b is $4c$

Now that we have the general formula, the distance in the case of $(-1, 1)$ and $(1, 1)$ is $4 \cdot c = 4 \cdot \tanh^{-1}(\frac{1}{\sqrt{3}}) \approx 2.633$

This result of a distance of $4c$ is interesting when $b=0$ as it shows that the distance does not depend on the value of a . we will revisit this when we make a conjecture of the distance in this part for different points

for the gaussians $(-2, 2)$ and $(2, 2)$ we similarly find a, b, c from the equation using $t \in [-1, 1]$

$$(\mu(t), \sigma(t)) = (a \tanh(ct) + b, a \frac{\sqrt{2}}{2} \operatorname{sech}(ct))$$

similar to the previous scenario

$$a \tanh(-c) + b = -2$$

$$a \tanh(c) + b = 2$$

$\Downarrow$

$$-2a \tanh(c) = -4$$

$$a \tanh(c) = 2$$

$\swarrow \searrow$

$$b = 0 \quad \tanh(c) = \frac{2}{a}$$

we get $c = \tanh^{-1}\left(\frac{1}{\sqrt{3}}\right)$

$$a \frac{\sqrt{2}}{2} \operatorname{sech}(c) = 2$$

$$a \frac{\sqrt{2}}{2} \sqrt{1 - \tanh^2 c} = 2$$

$$a \frac{\sqrt{2}}{2} \sqrt{1 - \frac{4}{a^2}} = 2$$

$$a \frac{\sqrt{2}}{2} \cdot \frac{1}{a} \cdot \sqrt{a^2 - 4} = 2$$

$$\sqrt{a^2 - 4} = \frac{4}{\sqrt{2}}$$

$$a^2 - 4 = \frac{16}{2} = 8$$

$$a^2 = 12$$

$$a = \pm 2\sqrt{3}$$

for simplicity we use $a = 2\sqrt{3}$
and because $-2\sqrt{3}$ provides an invalid geodesic

Our curve now becomes

$$\left(2\sqrt{3} \left(\tanh \left(\tanh^{-1} \left(\frac{1}{\sqrt{3}} \right) t \right) \right), \sqrt{6} \operatorname{sech} \left(\tanh^{-1} \left(\frac{1}{\sqrt{3}} \right) t \right) \right)$$

Plots in Jupyter notebook

We have calculated the general formula for distance when the family of curves of the geodesic is of the form $(a \tanh(ct) + b, a \frac{\sqrt{2}}{2} \operatorname{sech}(ct))$ where $b = 0$ as $4C$ since we got the same value of $C (\tanh^{-1}(\frac{1}{\sqrt{3}}))$ similar to the previous case the distance is $4C = 4 \times \tanh^{-1}(\frac{1}{\sqrt{3}}) = 2.633$

We can conjecture another general rule about geodesic distances from our calculation of distance as $4C$ when $b = 0$. $b = 0$ happens when the mean of the points is equidistant from mean $\mu = 0$. in both the cases here that is the case. The conjecture is that if the means are equidistant and the standard deviation (σ) of the distribution is same, then any scaling factor applied to both of the μ and σ in both the points does not change the geodesic distance. This can be attributed to the fact that any equal scaling is accounted by the parameter a which is not part of the distance and C that is the part of the distance calculation does not change with equal scaling

Part (f)

for Gaussians $(-1, 1)$ and $(1, 1)$

we found the equation of the curve when we found out the a, b, c for

these set of points and a geodesic curve equation. Intuitively since our t is from $[-1, 1]$ our Fréchet mean should be at $t=0$ which is the midpoint of the geodesic connecting $(-1, 1)$ and $(1, 1)$

$$\text{let us find } (\mu(0), \sigma(0)) = \left(\sqrt{3} \tanh(0), \frac{\sqrt{6}}{2} \operatorname{sech}(0) \right) = \left(0, \frac{\sqrt{6}}{2} \right)$$

$\downarrow \qquad \qquad \qquad \downarrow$
 $0 \qquad \qquad \qquad 1$

Plot in Jupyter notebook

observations about the mean distribution

* The standard deviation of the mean distribution is greater than the two distributions, this is because of the fact that the geodesic connecting the two distributions takes a curved path with distributions of increasing standard deviations along it

* The mean of the distribution is the mean of the μ values of the two distributions, which is also consistent with the midpoint of the geodesic lying at the centre of the μ component of the two distributions